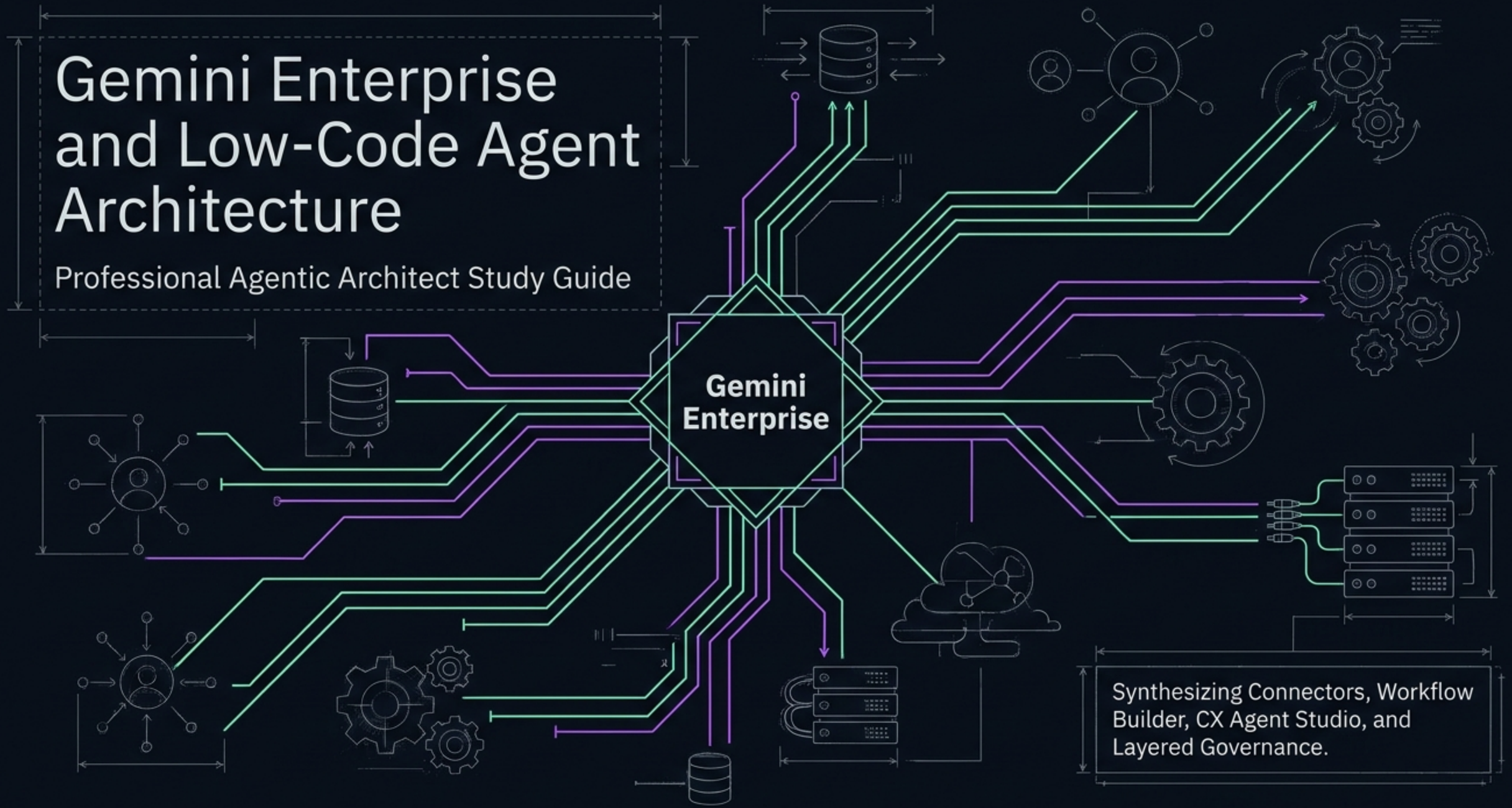
Gemini Enterprise and Low-Code Agent Architecture

Professional Agentic Architect Study Guide

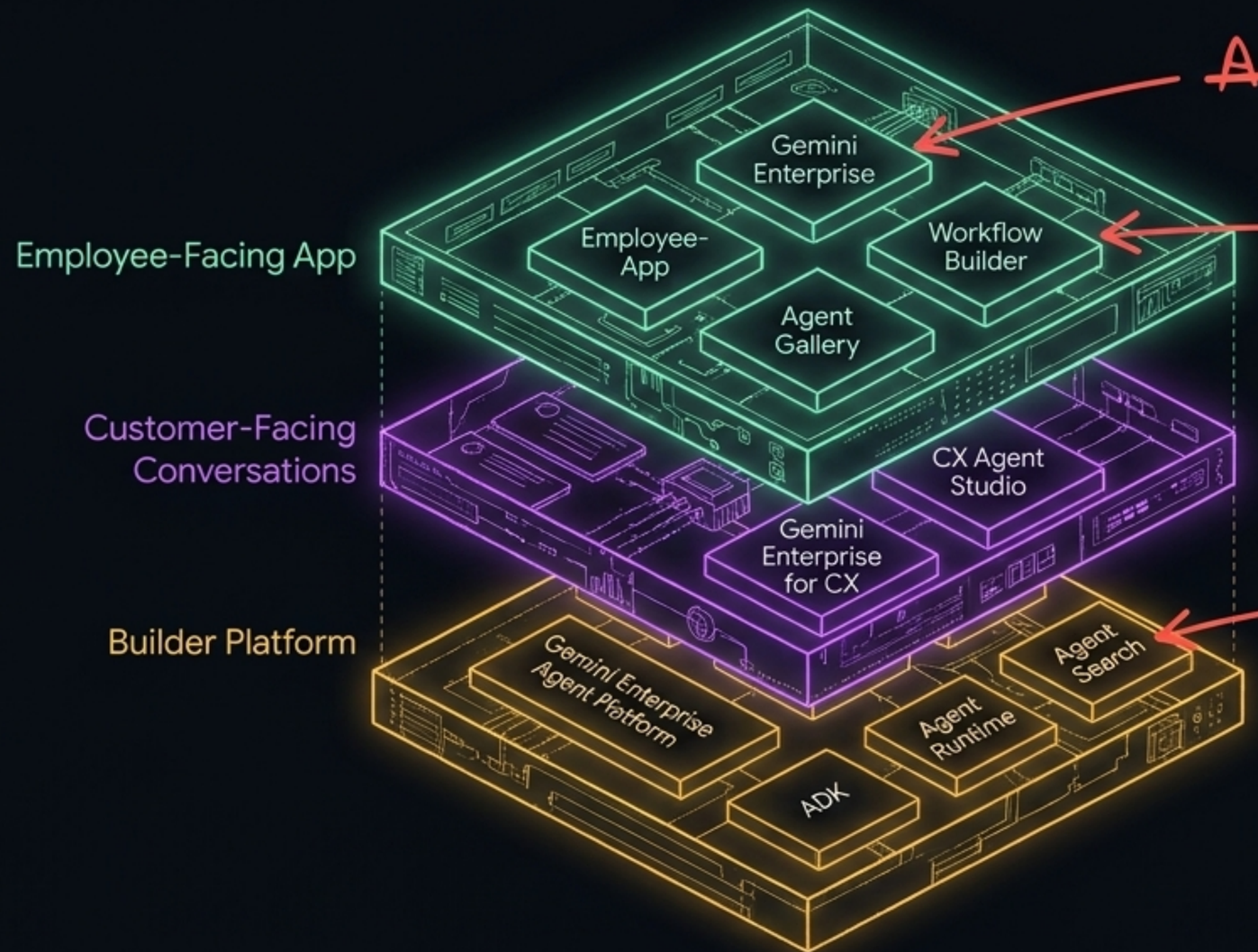


Triangulating the Audience for Agentic Platforms



Use Workflow Builder first for internal staff, CX Agent Studio for external consumers, and ADK when full control of code and runtime is required.

Defining the Architectural Layers and Resolving Name Drift

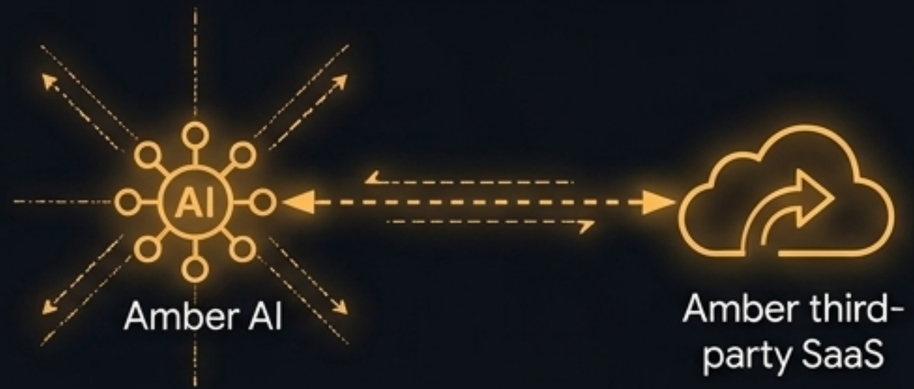


Edition Capabilities Dictate Platform Architecture Boundaries



Grounding Routing Trade-offs for Enterprise Data

Federated Search



- Live query to API
- Zero data copy
- Near-real-time

Trade-off: Subject to third-party rate limits and lower semantic ranking quality.

Data Ingestion

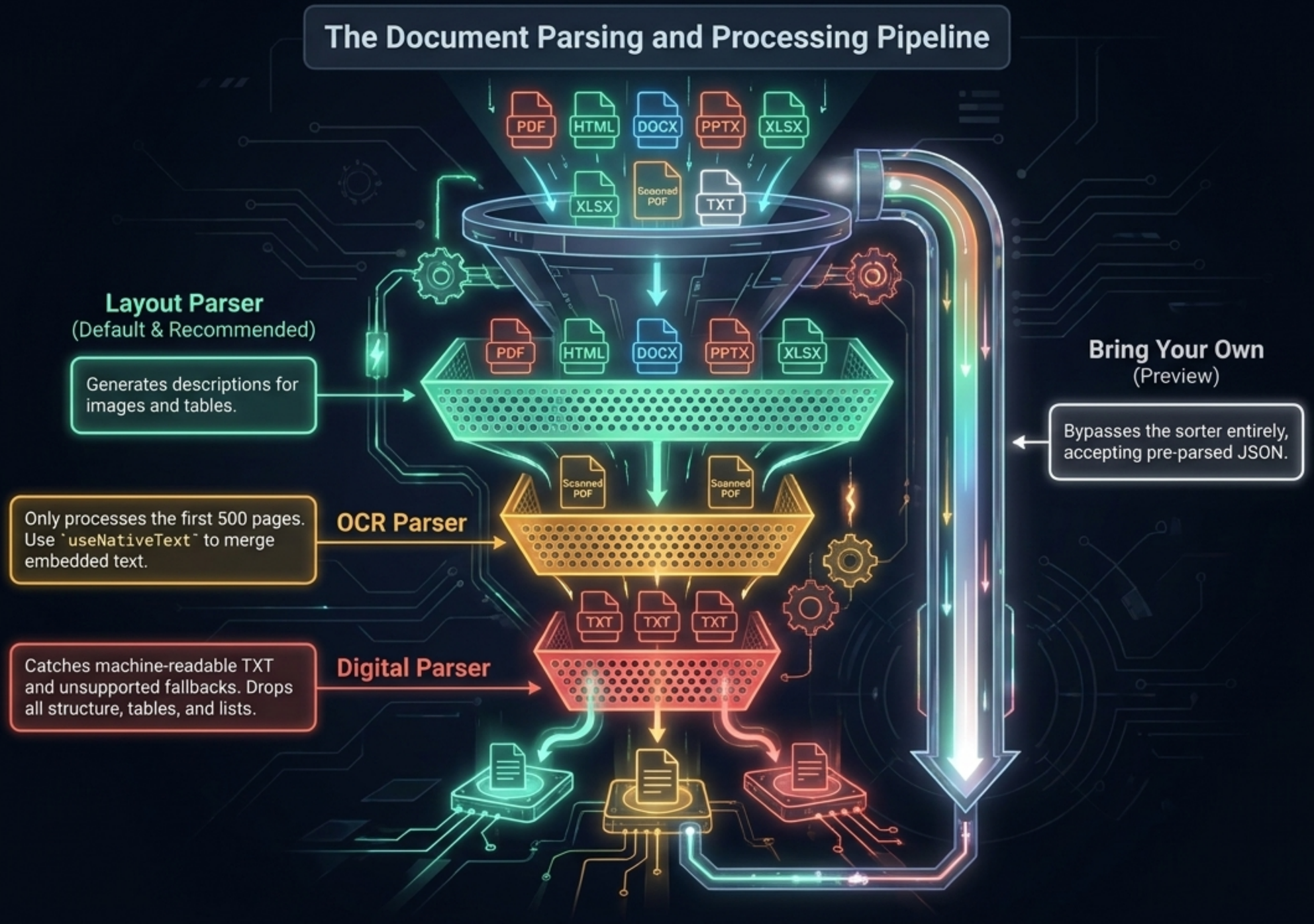


- Content and ACLs copied
- Indexed into Gemini Data Store

Trade-off: Best answer quality and semantic ranking, but requires storage quota and sync scheduling.

Exam Trap: Entity vs. Identity Sync

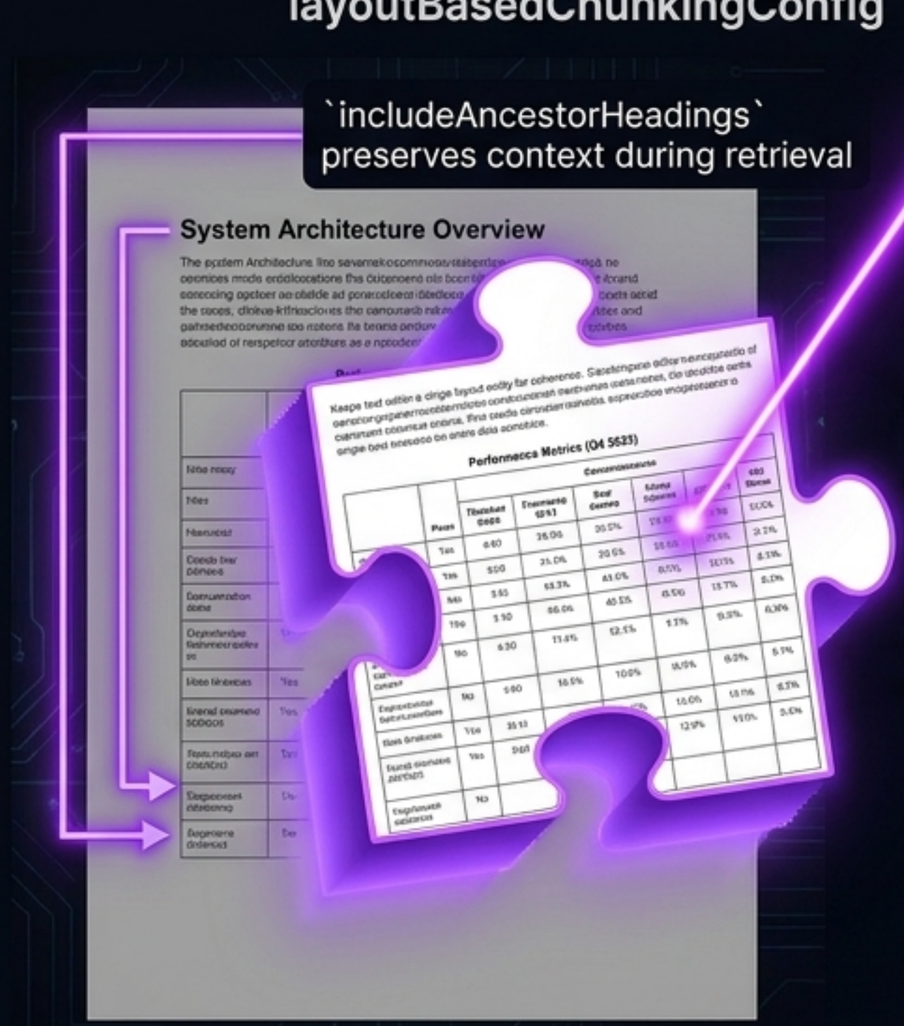
Both must run for ACL-correct answers (content records + group membership). Full syncs are strictly required to propagate deletions.


The Document Parsing and Processing Pipeline
PDF
HTML
DOCX
PPTX
XLSX
XLSX
Scanned PDF
TXT
Layout Parser
(Default & Recommended)
Generates descriptions for images and tables.
PDF
HTML
DOCX
PPTX
XLSX
Bring Your Own
(Preview)
Bypasses the sorter entirely, accepting pre-parsed JSON.
Scanned PDF
Scanned PDF
Only processes the first 500 pages. Use `useNativeText` to merge embedded text.
OCR Parser
TXT
TXT
TXT
Catches machine-readable TXT and unsupported fallbacks. Drops all structure, tables, and lists.
Digital Parser

Mechanics of Layout-Aware Chunking for Precise Retrieval

Contrasting precise, intelligent geometry against clumsy, rigid gridlines.

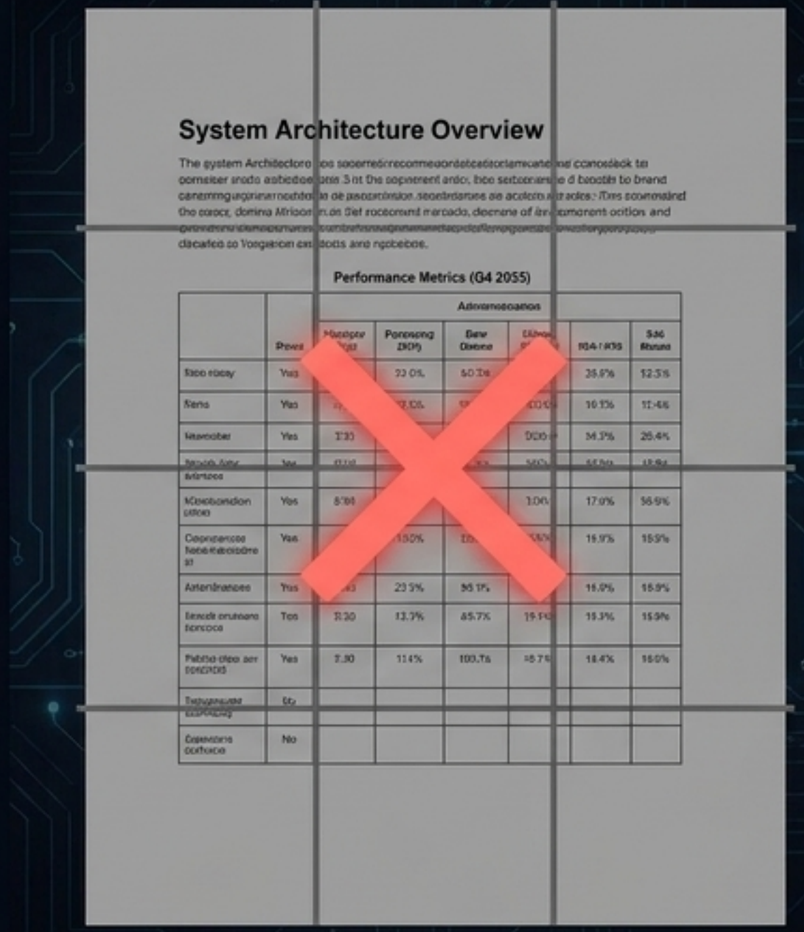
`layoutBasedChunkingConfig`



Keeps text within a single layout entity for coherence.

`chunkSize` constraint: 100 to 500 tokens (default 500).

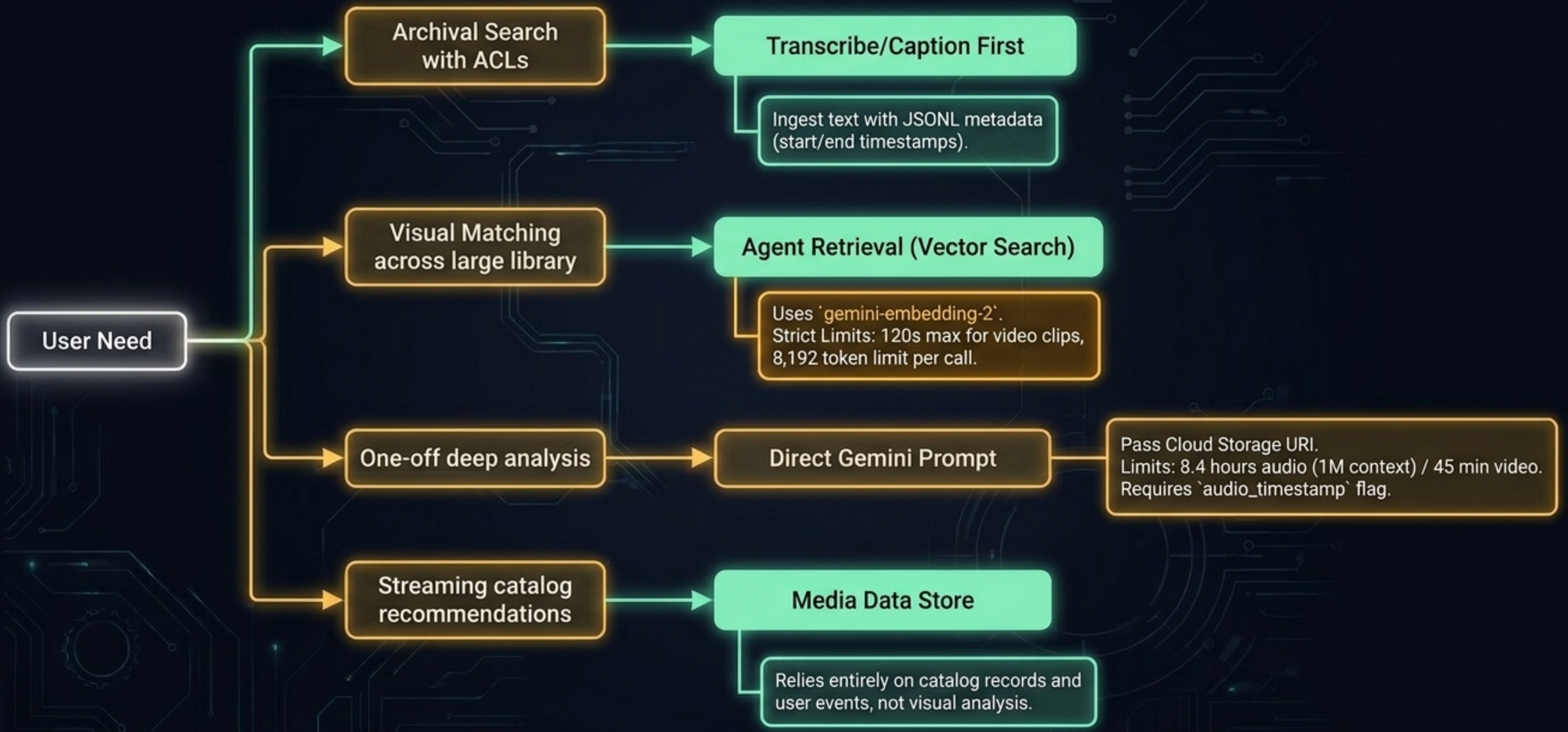
Arbitrary Gridline Chunking (Failure Case)



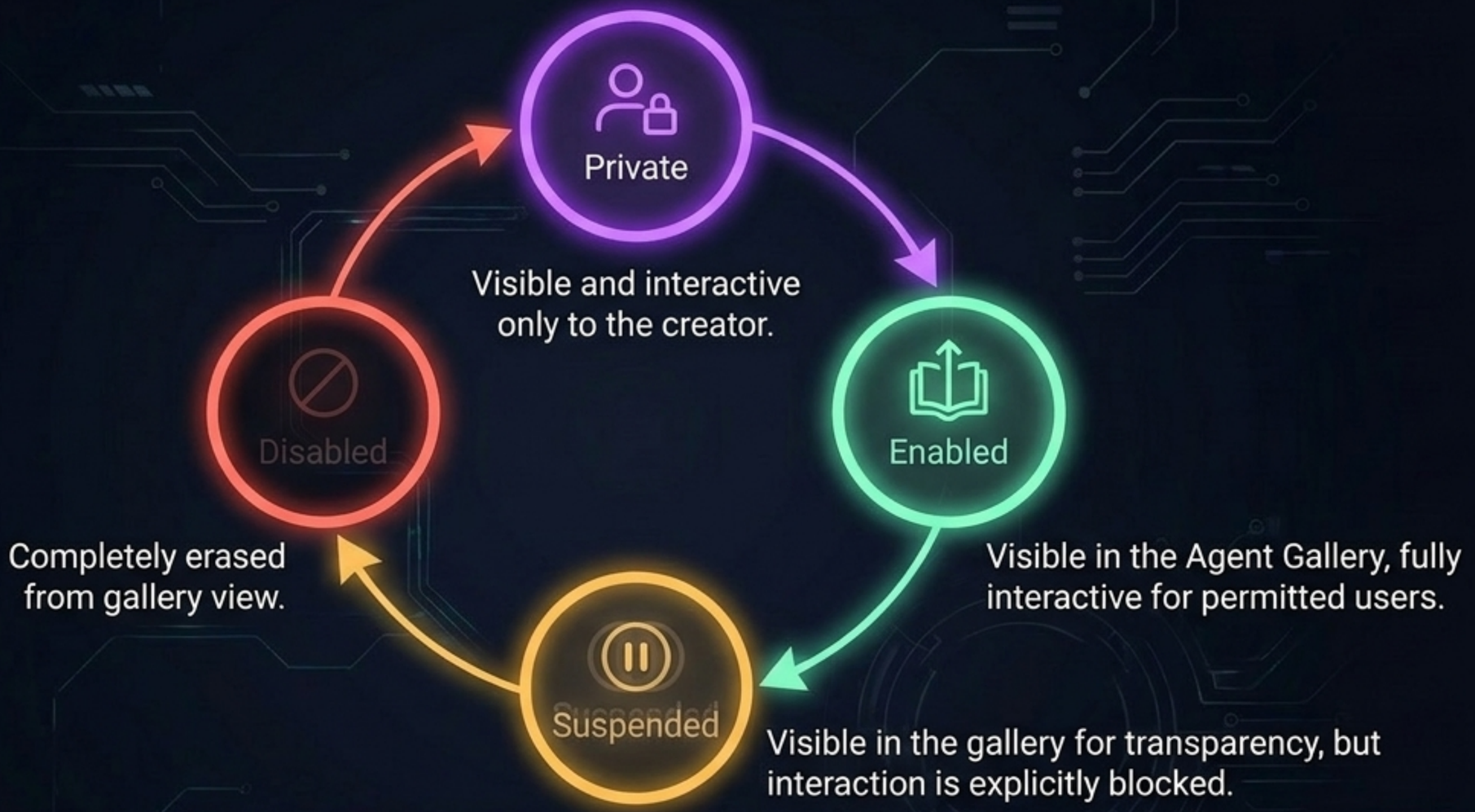
Arbitrary gridlines sever semantic coherence.

Tables and structures are broken, leading to context loss and poor retrieval quality.

Resolving Multimodal Constraints for Audio and Video



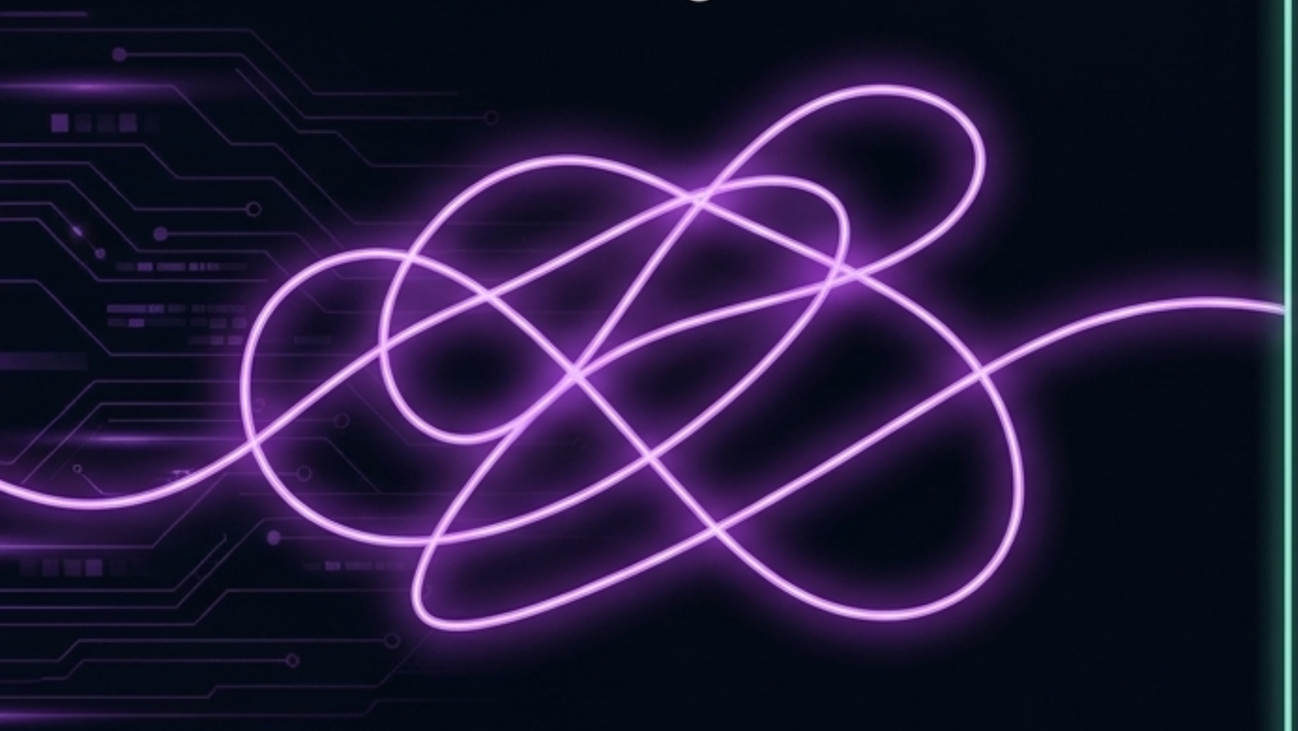
Agent Lifecycle States within the Internal Gallery



Agents are invoked via `@agent_name` in chat, or programmatically via StreamAssist API and A2A endpoints. Admins control sharing approvals and group availability.

Contrasting Conversational Agents and Sequential Workflows

Chat Agent



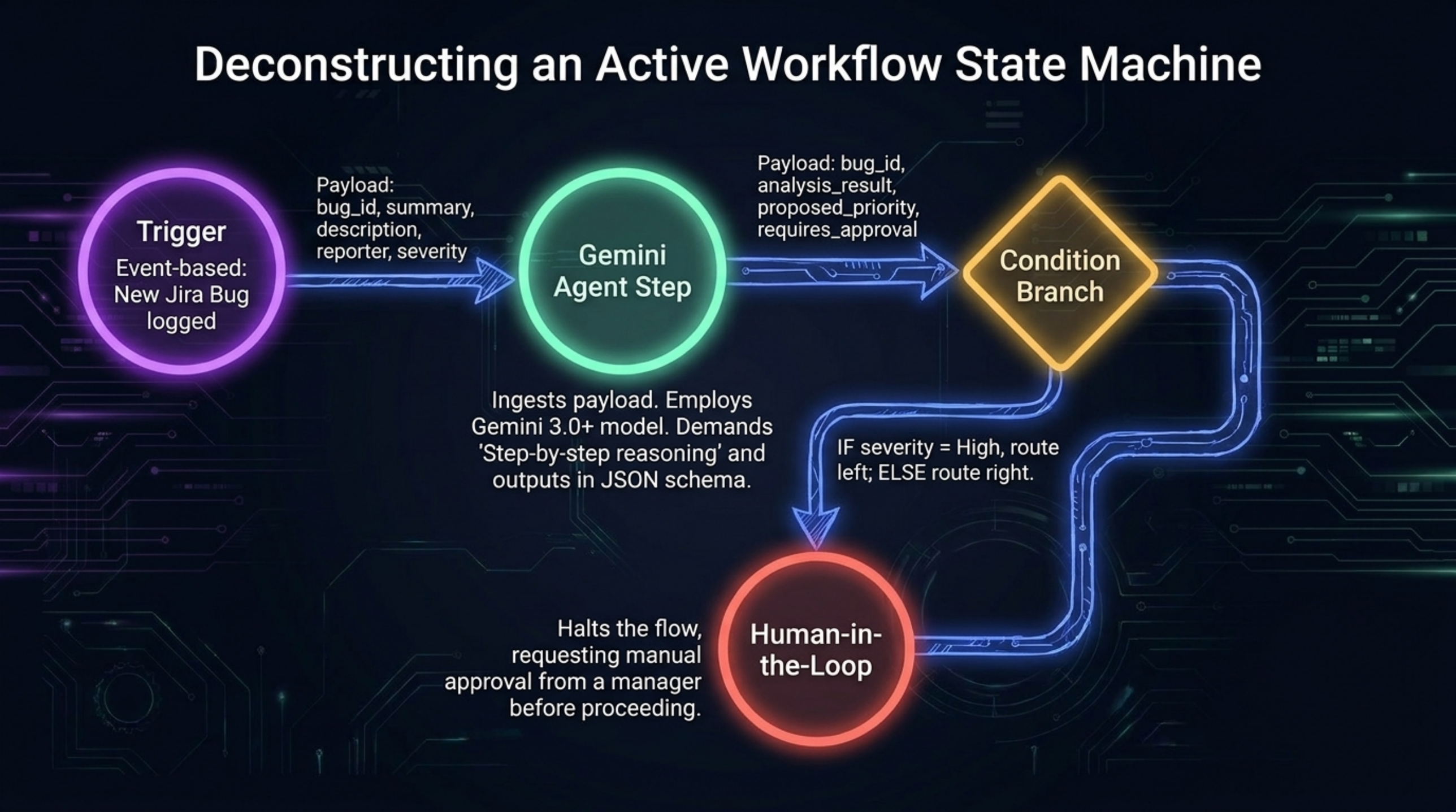
Non-deterministic. Started by a user `@mention`. Best for open Q&A, research helpers, and drafting. Context relies on prompt engineering and max 800k tokens of uploaded files.

Workflow



Ordered steps with passed variables. Triggered Manually, on a Schedule (cron), or by an Event. Best for repeatable triage, approvals, and report generation.

Deconstructing an Active Workflow State Machine



Connector Safeguards and the Human-in-the-Loop Imperative



Write actions strictly require a **Human-in-the-loop review** step to be executed.

Read Actions ("Search For Data")

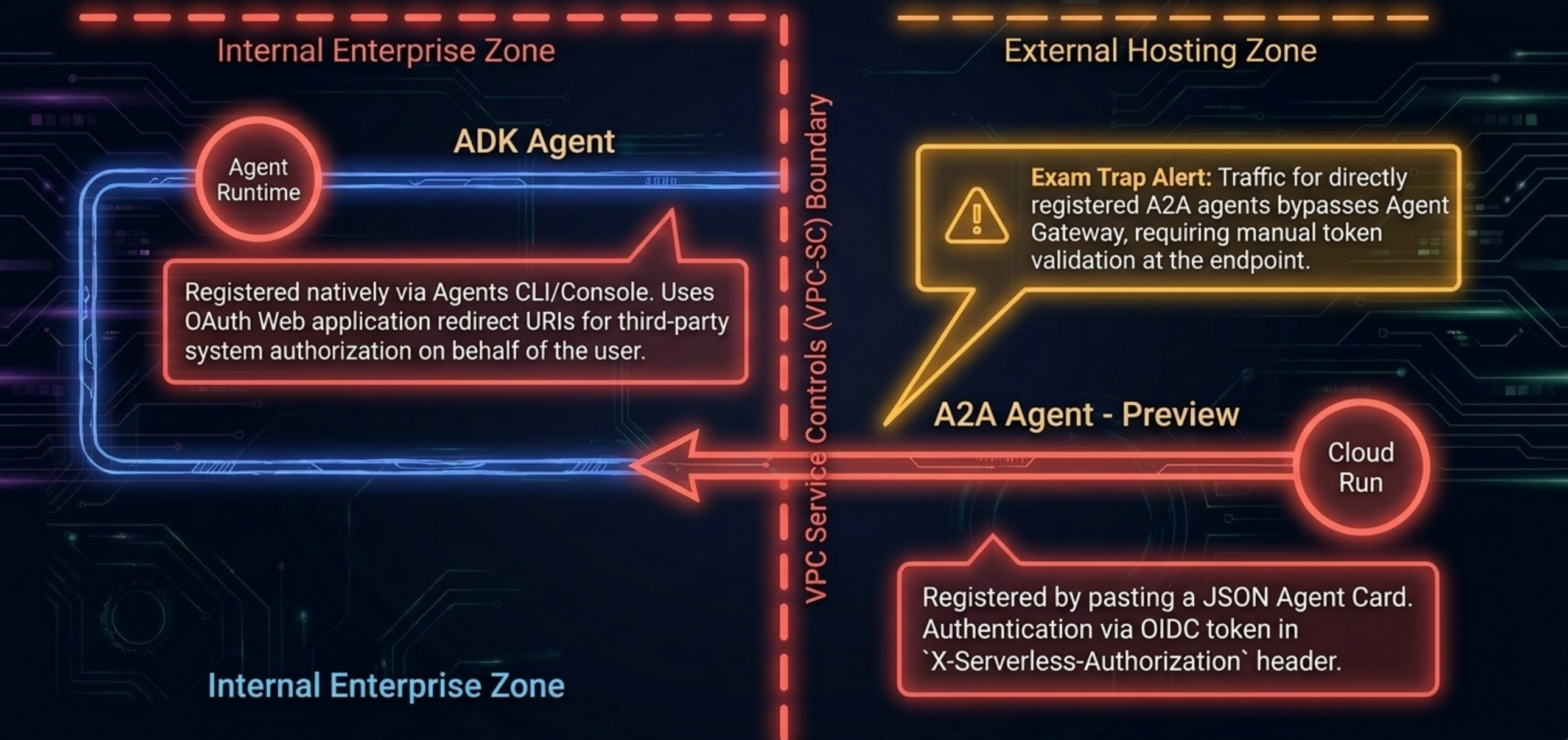
Granular, enabled per specific action. Evaluated against user access control lists at runtime.

Write Actions ("Add Or Update Data")

All-or-nothing toggle at the connector level.

Write actions will fail with `ENTITY_NOT_FOUND` if the corresponding Read actions for that connector are not also enabled.

Bridging the Gap: Integrating Custom ADK and A2A Agents



Architecting the Customer-Facing CX Application Tree

Root (Steering) Agent

Global Instructions

Attached to the Root Agent. Houses brand tone, DOs/DON'Ts. Inherited downward by all sub-agents.

structural prompting:
custer: ... primary_goal,
user: ... <taskfous.to...
...

<role>
<primary_goal>
<taskflow>
</step>

{@TOOL: name}

{@AGENT: name}

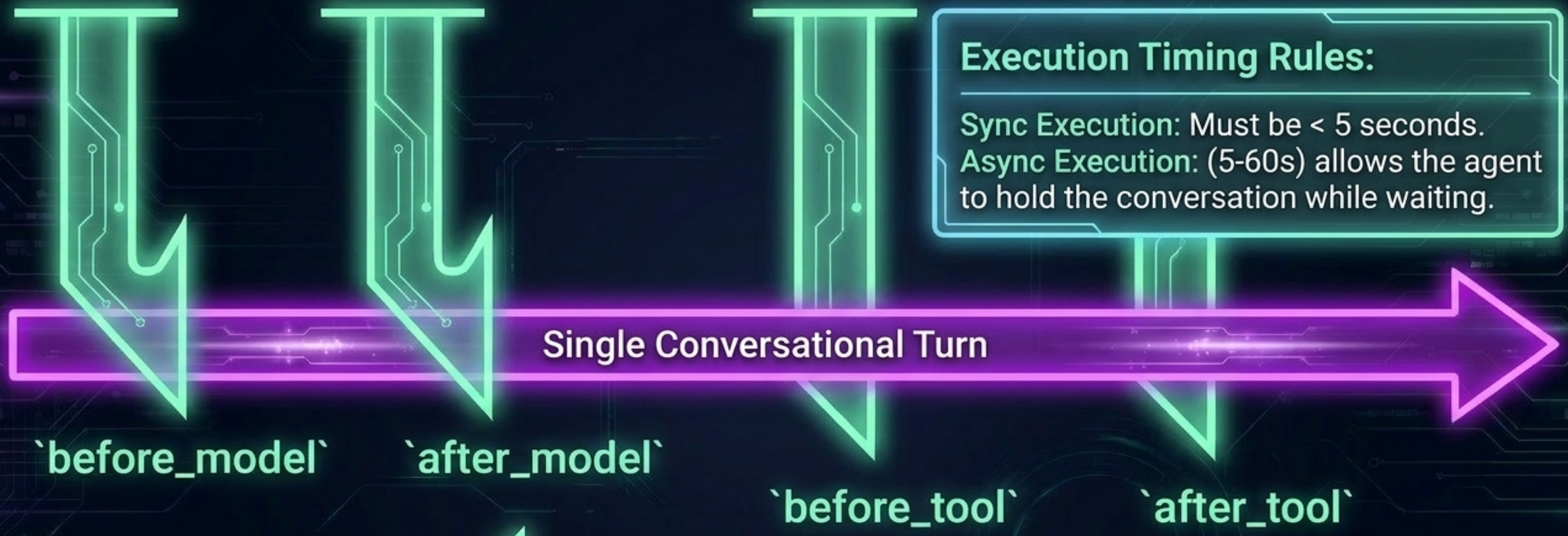
{username}

Specialized Sub-Agent

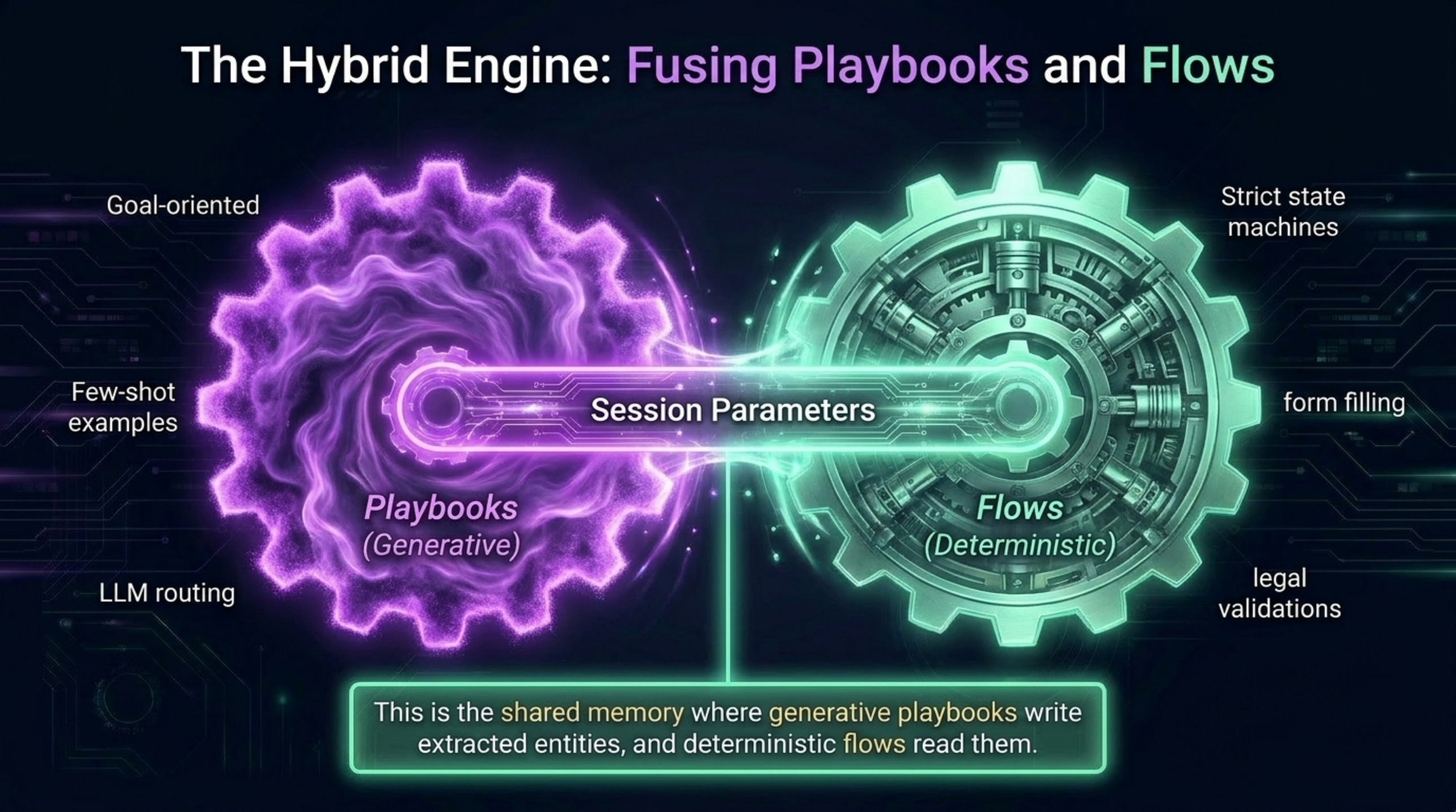
Specialized Sub-Agent

Specialized Sub-Agent

Tool Execution Timelines and Deterministic Callbacks

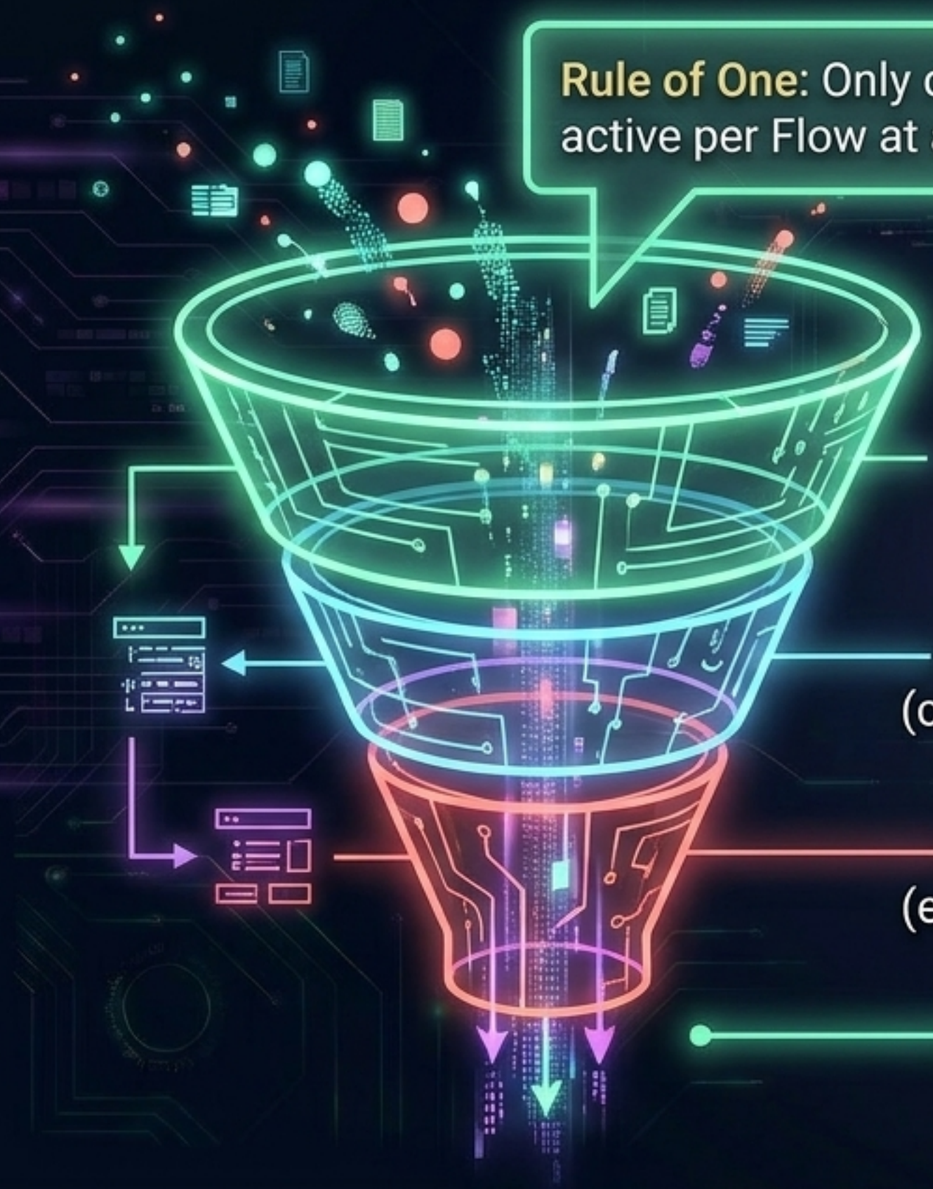


The Hybrid Engine: Fusing Playbooks and Flows



CX Page Anatomy and Turn-by-Turn Form Filling

Rule of One: Only one Page is active per Flow at any given time.



Entry Fulfillment
(fires first)

Form Filling
(collects variables)

Routes
(exit conditions)

Typed Extraction vs LLM Guessing

LLM Guessing

AI

Account#:
H4L-987654321

Hallucinated
& Unreliable

Typed Extraction

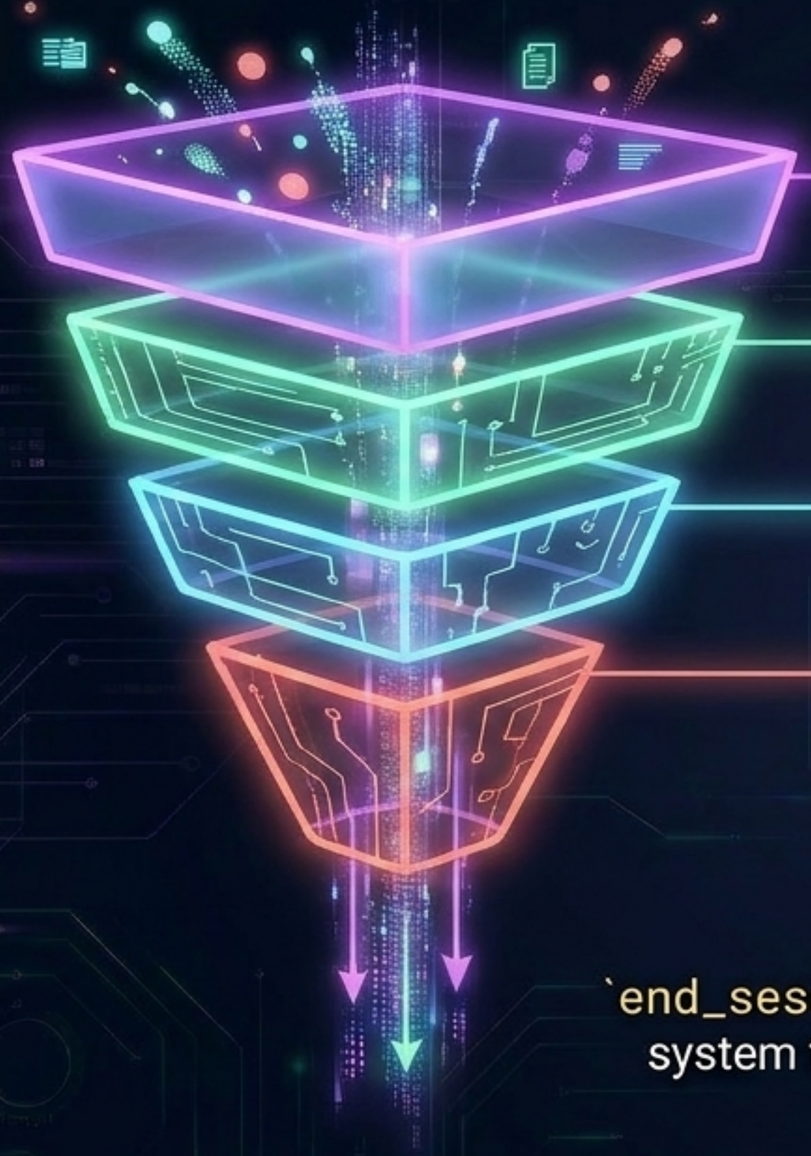
Form Parameter
bound
@sys.date

2023-10-27

Deterministic
& Type-Safe

Exit Condition: Routes trigger when `$page.params.status = "FINAL"`.

Handler Evaluation Hierarchies and Handoff Escapes



Scope Filter
Evaluates only active flow/page handlers.

Intent Routes
Evaluates named conversational intents.

Condition Routes
Evaluates system state logic.

Event Handlers
Catches fallbacks (`sys.no-match-1`, `webhook.error`).

The Stack Constraint

Flow-to-flow transitions push to a stack capped at exactly 25.

`end_session` system tool



⚠ Exam Trap

If using Agent Assist, `liveAgentHandoff.endSession` MUST equal `False`. Only set to `True` if bypassing Agent Assist completely.

Correcting State Machine Design Anti-Patterns

Silent Loops



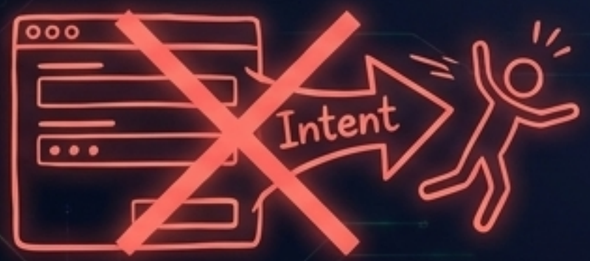
Fix: Add numbered event handlers (`no-match`/ `no-input`) escalating to human.

Stack Overflow



Fix: Add retry counters via session parameters and exit conditions to avoid crashing the 25-flow limit.

Broad Intents on Flows



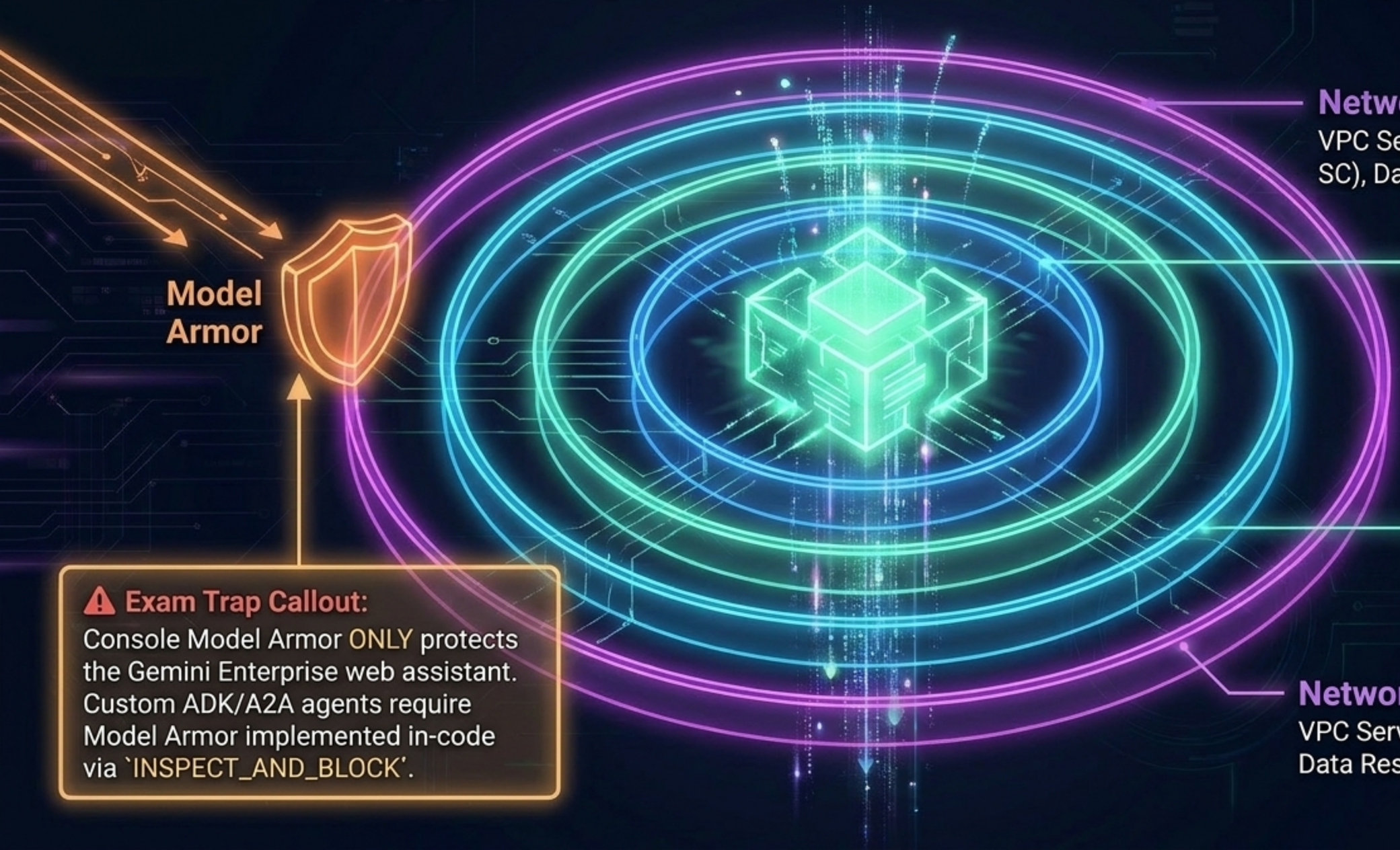
Fix: Scope intents strictly to the pages where they belong.

Reference Data in Instructions



Fix: Move FAQs to a Data Store tool; keep instructions strictly for behavior to save tokens.

Constructing the Layered Governance and Security Fortress



Network Layer
VPC Service Controls (VPC-SC), Data Residency, CMEK.

Data Layer
Source-system ACL enforcement via Admin connector provisioning, Gallery Sharing approvals.

Traffic Layer
Agent Gateway / OAuth proxying.

Network Layer
VPC Service Controls (VPC-SC), Data Residency, CMEK.

⚠ Exam Trap Callout:
Console Model Armor ONLY protects the Gemini Enterprise web assistant. Custom ADK/A2A agents require Model Armor implemented in-code via `INSPECT_AND_BLOCK'.

The Ultimate Low-Code vs. Pro-Code Architectural Matrix

	Workflow Builder	CX Agent Studio	Agent Studio	ADK on Agent Runtime
Audience	Employees	External Customers	Developers Prototyping	Dev/Platform Teams
Control Flow	Chat/Steps + HitL	Root + Sub-Agents + + Flows	Prompts	Full Custom (Loops/Parallel)
Grounding	Connectors & Files	Data Store/Web Search	RAG Engine	MCP, Agent Search, Custom
Evaluation	Execution Logs	Scenario, Simulation, Golden tests	Side-by-side	Custom Eval Suites

✓ Edition Checked. ✓ Data Mode Deliberate. ✓ Guardrails Armed. ✓ Architecture Validated.